

MINAMITORISHIMA, Japan

WMO: 479910

Lat: 24.290N

Lon: 153.979E

Elev: 7

StdP: 101.24

Time Zone: 9.00 (E09)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	18.1	18.8	9.1	7.2	19.2	10.1	7.7	19.6	12.3	21.1	11.5	21.0	5.9	10	0.492

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	4.2	31.8	26.4	31.4	26.3	31.0	26.2	27.5	30.2	27.1	30.0	26.8	29.7	4.6	110

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
26.7	22.3	29.1	26.3	21.8	29.1	26.0	21.4	29.0	87.0	30.3	85.4	30.0	84.2	29.8	29.1

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 12.2	11.0	10.0	DB	16.7	32.8	0.9	0.7	16.1	33.3	15.5	33.7	15.0	34.1	14.3	34.6	(4)
(5)			WB	12.2	27.9	0.8	0.5	11.6	28.3	11.1	28.6	10.6	28.9	10.0	29.3	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	26.0	22.4	21.9	22.8	24.6	26.3	28.2	28.6	28.5	28.6	28.1	26.6	24.7	(6)
(7)		DBStd	2.80	1.55	1.69	1.80	1.67	1.42	1.08	1.03	1.04	0.96	0.85	1.14	1.55	(7)
(8)		HDD10.0	0	0	0	0	0	0	0	0	0	0	0	0	0	(8)
(9)		HDD18.3	0	0	0	0	0	0	0	0	0	0	0	0	0	(9)
(10)		CDD10.0	5831	385	334	396	439	507	547	577	573	558	561	499	454	(10)
(11)		CDD18.3	2789	127	101	138	189	248	297	318	315	308	303	249	196	(11)
(12)		CDH23.3	25352	260	192	393	1024	2077	3357	3818	3753	3681	3414	2264	1118	(12)
(13)	CDH26.7	7145	4	4	21	116	398	1104	1401	1347	1328	1008	344	71	(13)	
(14)	Wind	WSAvg	5.4	5.7	5.5	5.7	5.6	5.0	4.1	4.7	5.1	5.7	6.0	6.1	5.8	(14)
(15)	Precipitation	PrecAvg	1044	71	44	63	54	74	50	155	157	93	109	70	94	(15)
(16)		PrecMax	1773	198	163	278	178	250	182	351	504	238	343	199	265	(16)
(17)		PrecMin	494	14	7	2	1	3	2	13	30	26	9	12	11	(17)
(18)		PrecStd	222	37	34	53	44	58	41	87	107	51	87	52	67	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	27.3	27.3	28.2	29.7	31.2	32.2	32.7	32.2	32.0	31.5	30.2	28.9	(19)
MCWB			23.2	23.0	23.6	25.1	25.8	26.2	26.8	26.6	26.8	26.3	25.4	24.8	(20)	
2%		DB	26.3	26.3	27.2	28.8	30.3	31.5	31.9	31.6	31.4	30.7	29.5	28.1	(21)	
		MCWB	22.4	22.3	23.1	24.5	25.5	26.0	26.4	26.5	26.5	26.0	25.0	24.2	(22)	
5%		DB	25.5	25.4	26.4	28.0	29.6	31.0	31.3	31.0	30.8	30.2	29.0	27.5	(23)	
		MCWB	21.7	21.6	22.5	24.0	25.2	25.9	26.2	26.4	26.3	25.8	24.7	23.7	(24)	
10%		DB	24.7	24.5	25.5	27.2	28.8	30.4	30.7	30.5	30.4	29.7	28.4	26.8	(25)	
		MCWB	21.1	20.9	21.9	23.5	24.8	25.7	26.0	26.2	26.2	25.5	24.4	23.2	(26)	
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	24.3	24.0	24.4	25.7	26.7	27.5	27.7	27.9	27.7	27.4	27.4	25.6	(27)
MCD B			26.3	26.0	27.1	28.7	29.7	30.5	31.2	30.4	30.5	29.9	28.6	27.9	(28)	
2%		WB	23.4	23.1	23.6	25.1	26.1	27.0	27.2	27.4	27.2	26.9	26.3	25.0	(29)	
		MCD B	25.4	25.3	26.3	27.9	29.2	30.2	30.5	30.0	30.0	29.3	28.3	27.1	(30)	
5%		WB	22.7	22.4	23.1	24.6	25.7	26.6	26.8	27.0	26.9	26.5	25.6	24.5	(31)	
		MCD B	24.8	24.7	25.6	27.1	28.5	29.7	30.0	29.8	29.8	29.1	28.0	26.7	(32)	
10%		WB	21.9	21.7	22.5	24.1	25.2	26.2	26.5	26.6	26.6	26.2	25.1	24.0	(33)	
		MCD B	24.4	24.0	24.9	26.4	27.9	29.3	29.7	29.5	29.5	28.8	27.7	26.3	(34)	
(35)	Mean Daily Temperature Range	5% DB	MDBR	3.5	3.9	4.0	4.0	4.1	4.3	4.2	4.0	3.6	3.5	3.3	3.3	(35)
MCDBR			4.2	4.8	5.0	4.9	5.0	4.7	4.8	4.4	4.0	3.8	3.7	3.8	(36)	
MCWBR			3.3	3.3	3.0	2.5	2.3	2.1	2.2	2.2	2.1	2.0	2.1	2.3	(37)	
5% WB		MCD BR	3.9	4.5	4.8	4.7	4.7	4.5	4.5	4.2	3.8	3.6	3.5	3.7	(38)	
		MCWBR	3.6	3.5	3.1	2.6	2.4	2.2	2.3	2.3	2.1	2.0	2.2	2.5	(39)	
(40)	Clear-Sky Solar Irradiance	taub	0.381	0.384	0.411	0.432	0.420	0.397	0.417	0.416	0.415	0.417	0.395	0.389	(40)	
taud		2.520	2.492	2.423	2.416	2.487	2.567	2.517	2.522	2.519	2.490	2.537	2.524	(41)		
Ebn at Noon		878	901	889	873	876	890	872	876	873	856	858	855	(42)		
Edh at Noon		95	104	116	119	111	102	107	106	104	103	93	92	(43)		
(44)	All-Sky Solar Radiation	RadAvg	3.72	4.66	5.48	6.23	6.60	7.14	6.38	5.89	5.87	5.19	4.30	3.60	(44)	
(45)		RadStd	0.25	0.28	0.31	0.47	0.52	0.34	0.42	0.68	0.38	0.20	0.21	0.20	(45)	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)		
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (0 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page